

KUSHAGAR SINGH AHUJA

Software Development Engineer
(Full-Stack / ML)



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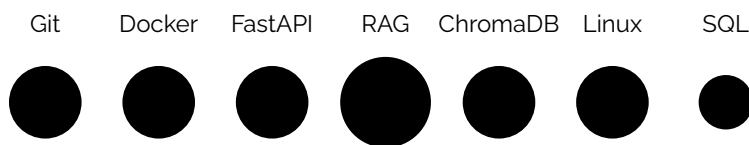
LinkedIn



Portfolio

WHO AM I?

Entry-level **Software Development Engineer / AI Engineer** with experience in **LLM systems, Retrieval-Augmented Generation (RAG), backend engineering, and full-stack development**. Built AI pipelines using **Python, FastAPI, ChromaDB, Docker, and LLM APIs** with focus on **semantic retrieval, vector search, reranking, and scalable technical QA systems**.



EXPERIENCE

Feb 2025 – Present **AI / ML Engineer Intern**

Nokia

- Working on enterprise-scale **LLM and RAG systems** for technical document intelligence and QA workflows.
- Developing retrieval pipelines using **semantic search, reranking, metadata-aware indexing, and vector databases**.
- Contributing to backend optimization, retrieval evaluation, and scalable AI infrastructure development.

PROJECTS

2025 – Present

Advanced Hybrid RAG System for Technical Document Intelligence

Python · FastAPI · ChromaDB · LLMs · Docker

- Built a hybrid **Retrieval-Augmented Generation (RAG)** pipeline using semantic retrieval, hierarchical chunking, reranking, and query routing.
- Implemented structured PDF ingestion, metadata-aware indexing, and adaptive retrieval workflows using ChromaDB and LLM APIs.
- Integrated caching, fallback retrieval, and context compression to improve retrieval relevance, scalability, and response accuracy.

2025

AI-Assisted Flood Risk Intelligence & River Discharge Forecasting
Time-Series Forecasting

PyTorch · Deep Learning ·

- Developed a multivariate time-series forecasting pipeline for hydrological discharge prediction using PyTorch.
- Implemented hybrid **BiGRU + Attention** architectures for temporal sequence learning and forecasting optimization.
- Evaluated model performance using **MAE, RMSE, MAPE, and R²** for flood monitoring scenarios.

2025	Synthetic Dataset Generation using Diffusion Models	PyTorch · Stable Diffusion · LoRA
	• Built a parameter-efficient LoRA fine-tuning pipeline for Stable Diffusion-based synthetic biometric image generation. • Integrated preprocessing, latent caching, TensorBoard logging, and EMA optimization for scalable training workflows. • Worked on GPU-efficient experimentation and diffusion-model fine-tuning pipelines.	

EDUCATION

2022 – 2026	B.Tech, Computer Science & Engineering CGPA: 7.06 / 10 (current)	SRM Institute of Science and Technology, Ramapuram
2020 – 2022	Class XII Percentage: 82%	Career Point Gurukul, Mohali, Chandigarh

CERTIFICATIONS

2025	Oracle Cloud Infrastructure – foundational certification Completed an Oracle Cloud Infrastructure certification covering core cloud concepts, compute, storage and networking services, and basics of deploying applications on OCI.	Oracle
2024	Building AI-Powered Search with MongoDB Vector Search Learned how to use vector embeddings and MongoDB Vector Search to implement semantic search and AI-augmented querying in application backends.	MongoDB University

RESEARCH PUBLICATION

2026	A Real-Time AI-Assisted Latent Data Assimilation Framework for Flood Risk Intelligence and Hydrological Monitoring	ICACCT'26 — Francis Xavier Engineering College
	• Published and presented research on deep learning-based hydrological forecasting and flood-risk intelligence. • Implemented and evaluated recurrent architectures including LSTM, BiGRU, and Attention-based models for multivariate sequence prediction. • Worked on preprocessing pipelines, temporal feature extraction, and regression-based forecasting evaluation.	